Coding in Terminal Guide

Checkout the video!
By Orla Collins

Why?

- Can allow you to access files faster than a Graphical User Interface (GUI)
- Is often used with Git/Github for version control of projects
- Can be helpful when programs like Jupyter Notebook aren't running, *if* you have the files saved to your computer

How?

- Your computer likely has a preinstalled version of a terminal or command line interface
  - In Apple, this is Terminal
  - In Windows, it may be Command Prompt or PowerShell
  - In Linux, it may be GNOME Terminal
- With some relatively basic commands, you can navigate through and run files on your computer. This is something that computer scientists are expected to understand.

File Access Commands

- `ls` (or `dir` on Windows): lists files or directories in your current location
- `cd [directory_name]`: **c**hanges the **d**irectory you're into your chosen one
- `mkdir [directory_name]`: makes a new directory (folder)
- `touch [file_path]`: make a new file
- `cp [source] [destination]`: copies a file/directory to the specified location
- `mv [source] [destination]`: move or rename a file/directory
- `rm [file_name]`: remove a file
- `cd ..`: back up a directory level
- `pwd`: check current directory
- `clear`: clear the Terminal window

Python in Terminal

- Can open many files in Terminal but .py files are particularly compatible
- `python [file_name.py]:` runs a Python file
- `python --version:` tells you the version of Python running on your device, if you are having issues you may want to update the version
- Writing text in terminal:
  - `echo [text]:` print text to terminal
  - `echo 'print("text")' > [file_name]:` print the text to a file
  - `echo 'print("text")' >> [file_name]:` *append* the text to a file

Installing Packages

- `python -m pip list:` lists all of the packages currently installed on your device
- `python -m pip install [package_name]:` install or reinstall a package

Interact with Git

- `git init:` initialize an empty Git repository
- `git status:` check the status of your repository
- `echo ["text"] > .gitignore:` tells Git to ignore the file or text
- `ls -a:` gives a description of everything in the repository
- `git add [file_name]:` add a file to your repository
  - `git add .:` add everything in your current directory
- `git commit -m ["text_explanation"]:` commit changes to repository with a description
- `git branch -M main:` connects to your GitHub repo main branch (or replace main with your branch of choice)

Getting a Github Token

- In order to connect your local Git repository to GitHub, you will need a personal access token
  - In GitHub, go to Settings, Developer Settings, Personal access tokens, Tokens (classic), select “repo” and generate
  - Save this token code on your computer
  - When `git push -u origin main` prompts for a password, paste your token

Pushing to GitHub

- `git remote add origin <web_link>:` connects to your online Github repository
- `git push -u origin main:` push changes from your local repo to GitHub